

Upgrading Medical Devices: A Review of Operational Remanufacturing Constraints

Abstract

The healthcare sector faces growing pressure to reduce environmental impact while maintaining safety, performance, and regulatory compliance. Remanufacturing offers a potential pathway to extend the lifetime of medical devices by restoring and upgrading their technical functionality, yet its practical implementation remains constrained. This literature review synthesizes findings from 33 academic and gray literature sources to examine the operational constraints shaping remanufacturing strategies for upgrading existing medical devices. The review identifies three interdependent constraint areas. First, product design and architecture determine whether devices can be safely disassembled, cleaned, upgraded, and reassembled across multiple lifecycles, with modularity, material durability, and component access emerging as main prerequisites. Second, business models and supply-chain structures influence the timing, quality, and availability of returned devices, affecting economic feasibility and process effectiveness. Third, regulatory, safety, and liability requirements strongly condition possible remanufacturing practices, with ambiguous definitions and hygiene standards acting as both barriers and safeguards. Overall, the findings indicate that upgrade-oriented remanufacturing is feasible only when design decisions, return logistics, and regulatory frameworks are correctly aligned, highlighting the need for healthcare-specific frameworks that integrate technical, operational, and compliance considerations.

Keywords

Remanufacturing, Medical device, Design for upgrade, Circular economy, Supply chain

1 Introduction

The healthcare sector is responsible for about 5% of global greenhouse gas emissions (Or & Seppänen, 2024), of which 71% results from supply-chain factors such as production and transportation (Das & Chandra, 2022). While healthcare is concerned with *care* and *cure*, this negative impact on the climate contributes to the exact opposite by having major adverse consequences for human health (Rossati, 2017).

As such, it becomes clear that efforts need to be undertaken to reduce the climate impact of the healthcare sector. Whereas the top R-strategies (Refuse, Rethink, Reduce) focus on smarter use of fewer products, the subsequent strategies (Reuse, Repair, Refurbish, Remanufacture, Repurpose) center on extending the lifecycle of the products that *do* end up being used (Ellen MacArthur Foundation, n.d.).

As medical devices are subjected to stringent quality norms, remanufacturing is often the most feasible way to retain value in an eco-efficient manner, as "other R-strategies result only in marginal returns or even none at all" (Schulte et al., 2021). Remanufacturing follows a structured, indus-

trial process in which used devices are systematically collected, inspected, disassembled, cleaned, selectively repaired or upgraded, reassembled, and tested to meet the same regulatory and functional requirements as newly manufactured products.

Only once the useful lifecycle of a product can no longer be extended should the final R-strategies (Recycle, Recover) be used to reclaim raw materials or embedded energy.

One of the primary challenges to extending medical device lifecycles is technological obsolescence, where devices remain physically functional but become incompatible with evolving technologies or clinical requirements. At this stage, remanufacturing may be employed not only to restore functionality, but also to upgrade devices so that their specifications exceed those of the original production (Asif et al., 2021).

As care and cure strategies continuously change and improve, these changes may be bundled with circular design strategies to create medical devices that are upgraded over time, while simultaneously causing less overall environmental impact than manufacturing new devices (Leemans et al., 2025; Schulte et al., 2021).

2 Method

To inventory the field of remanufacturing of medical devices, this literature review aims to answer the following question:

What operational constraints are documented in remanufacturing strategies for upgrading technical functionality in existing medical devices?

This literature review is based on 33 selected works. The literature review protocol, including the search strings and retrieval process, can be found in [Appendix A](#). Inclusions were restricted to articles, conference proceedings, and gray literature published after 2005, limited to English or Dutch language, and using titles-keywords-abstracts as search fields.

ChatGPT 5.1 was used to generate a set of comprehensive and MECE (Mutually Exclusive, Collectively Exhaustive) search queries for Google Scholar, with explicit instructions to apply strategic wildcards. The resulting search strings were arranged from broad to increasingly specific:

1. ("medical device*" AND (retrofit* OR remanufactur* OR "design modific*" OR "post-market upgrade*")) AND ("safety-critical" OR "risk control*" OR "performance improvement") AND (MDR OR FDA OR "ISO 14971" OR "ISO 13485" OR "regulatory constraint*" OR "technical feasibility")
2. "medical device*" AND (retrofit* OR upgrade*) AND ("ISO 13485" OR "ISO 14971" OR IEC*)
3. "medical device*" AND ("design change*" OR modific*) AND (MDR OR "EU 2017/745" OR "Medical Device Regulation")
4. "medical device*" AND ("design change*" OR modific*) AND (FDA OR 510(k) OR "Quality System Regulation")
5. "medical device remanufacturing" AND (MDR OR "Medical Device Regulation" OR FDA OR "regulatory compliance")
6. "medical device*" AND remanufactur* AND ("lifecycle" OR "circular" OR "reuse")
7. "medical device*" AND (retrofit* OR "design change*") AND ("human factor*" OR usability OR workflow)
8. (retrofit* OR upgrade*) AND ("legacy system*" OR "embedded system*" OR "sensor integration") AND "medical device"
9. ("sensor*" OR "identification technology*" OR "barcode*" OR RFID OR "computer vision") AND ("medical device*" AND retrofit*)
10. "risk control*" AND ("medical device*" OR "health technology") AND (upgrade* OR modific*)
11. ("safety-critical" OR "safety function*") AND ("medical device*" OR "health-care equipment") AND (upgrade* OR retrofit*)
12. "post-market" AND ("modification*" OR upgrade* OR retrofit*) AND "medical device"
13. "medical device*" AND retrofit* AND ("upgrade*" OR "moderni*ation" OR "function* improvement")
14. "medical device*" AND retrofit* AND ("safety-critical" OR "safety function*" OR "risk control*")
15. remanufactur* AND "medical device*" AND ("safety" OR "performance upgrade*" OR "risk reduction")
16. (retrofit* OR "field upgrade*") AND ("automated medication" OR "drug dispens*" OR "medicine dispens*" OR "health* automation")

This results are structured into three main subsections. [Section 3.1](#) delves into design and product architecture constraints that affect repeated lifecycles and component replacement. [Section 3.2](#) covers business-model and supply-chain constraints that affect product core availability and process stability. [Section 3.3](#) outlines safety, liability, and regulatory constraints that define allowable practices and required guardrails.

3 Results

3.1 (Designing for) Multiple Lifecycles

A frequently reported constraint for remanufacturing is whether a device can be opened safely and non-destructively. Sundin (2004, as cited in Gray and Charter, 2007) identifies ease of access, part identification, wear resistance, and handling as critical factors, complemented by cleanability (Kane et al., 2017) and the reduction of hazardous components (Baron, 2016; Hoveling et al., 2024). Several authors emphasize addressing these issues *during* design; for example, Leemans et al. (2025) recommended a disassembly analysis to ensure critical components can be accessed or disassembled properly. They describe disassembly access, cleanability, and material and chemical considerations as conditions shaping whether remanufacturing can be carried out within safety and cost boundaries.

Relatedly, *Design for Disassembly* is presented as enabling part removal without damaging the product, thereby allowing replacement without harming other components (Ahlstedt & Sundin, 2023; Barquet et al., 2013; Gray & Charter, 2007). The reviewed literature notes implications for third-party remanufacturers, who depend on designs that reduce accidental breakage, while original equipment manufacturers (OEMs) may be reluctant to cooperate as they do not want to share product lifetime information (Fofou et al., 2021). Constraints also arise from corrosion or material deterioration, which can successful component liberation (Gray & Charter, 2007; Ijomah et al., 2006). In response, durable materials, reversible connections, and standardized parts are emphasized (Hatcher et al., 2014, as cited in Kane et al., 2017). Across these sources, fastener types, materials, and part standardization are repeatedly linked to whether disassembly and replacement can be executed without disproportionate risk, time, or scrap.

Several studies (see Table 1) describe analytical tools to identify redesign opportunities that improve remanufacturability. These tools are presented as structured means to evaluate components, processes, and redesign priorities. Variation in how these tools are used indicate that remanufacturing evaluation practices are not standardized across contexts or product types.

Several authors note that existing methods often emphasize economic or environmental performance while rarely social impacts (Leemans et al., 2025; Mejía-Moncayo et al., 2023).

Another reported constraint concerns component survival across repeated use–remanufacturing–use cycles, influencing whether replacement, retrofitting, or upgrading remains technically and economically feasible. A lack of methodological guidance for designing multi-lifecycle products is described as a continuing challenge (Asif et al., 2021; Samenjo et al., 2023). Debate remains for certain medical devices; while some argue that single-use medical devices should *not* be refurbished (Plumeyer & Braun, 2011), others report successful remanufacturing of items such as single-use electrophysiology catheters (Guo et al., 2014; Schulte et al., 2021). Local legislation is also reported to increase administrative burden (Benedettini, 2022). For medical devices, feasibility is described as dependent on technical characteristics (materials, interfaces, deterioration) and governance conditions (device classification and jurisdiction-specific requirements).

Residual value consistently emerges as a key determinant of whether remanufacturing is pursued and which strategies are viable, closely tied to the timing and condition of product return (Sierra-Fontalvo et al., 2024; Vogt Duberg et al., 2023). OEM-planned lifetimes may be shorter than actual use phases; products must be recovered while they remain technically workable and economically valuable (Plumeyer & Braun, 2011). To address this, predefined return cycles have been proposed, enabling restoration to original or upgraded specifications and aligning remanufacturing with Design for Upgrade and modular design principles (Asif et al., 2021; Gray & Charter, 2007). Modularity is important, as different forms of obsolescence occur at different rates, allowing selective component replacement while retaining the remaining system (Asif et al., 2021). Platform-based modularization further supports component reuse across product families and generations, expanding the scope for reuse, specification changes, and technical improvements during remanufacturing (Gray & Charter, 2007; Ijomah et al., 2006; Kane et al., 2017).

Table 1

Analytical Tools to Identify Redesign Opportunities

Method	Authors	Application
RemPI	Sierra-Fontalvo et al. (2024)	Identify product redesign opportunities to improve remanufacturability
Product Profile Analysis	Zwolinski et al. (2006)	Study successfully remanufactured devices and consider technical and economic aspects
Fuzzy AHP	Chakraborty et al. (2017)	Evaluate product design alternatives and identify the critical remanufacturing step
Remanufacturability Index	James et al. (2021)	Identify product design attributes that influence remanufacturability
RemPro	Sundin and Bras (2004)	Identify which technical product property influences which remanufacturing step
Two-step Assessment	Ahlstedt and Sundin (2023)	Assess remanufacturability based on economic criteria and product design
5AFIR	Vogt Duberg et al. (2023)	Assess and establish prerequisites to initiate and industrialize remanufacturing
Medical Device Remanufacturing Methodology	Reinbold and Zaeh (2025)	Enable compliant, economically and environmentally sustainable remanufacturing of medical devices
Remanufacturing Sustainability Index	Steinhilper and Weiland (2015)	Assess remanufacturability on technological effort, remanufacturing yield, and usage patterns

3.2 Business Model and Supply Chain

A main operational constraint described across studies is securing enough cores (returned products) at the right time to maintain efficient rework operations ([Benedettini, 2022](#)). Closed-loop supply chain (CLSC) contains both forwards and reverse flows to retrieve cores ([Fofou et al., 2021](#)). Trade-in programs are described as mechanisms to support returns, with examples including Philips Refurbished Systems ([Pagh Jensen et al., 2019](#)) and Caterpillar ([Gray & Charter, 2007](#)). Such programs are ways to reduce uncertainty in core acquisition.

Product-service systems (PSS) are presented as another approach, enabling product return during the middle-of-life (MoL) before components become too worn ([Barquet et al., 2013](#); [Fofou et al., 2021](#)). In contrast, low-resource settings may use products until failure due to scarcity ([Otur et al., 2021](#)). Because PSS models incorporate reverse logistics, they are argued to achieve higher return rates ([Benedettini, 2022](#); [Ijomah et al., 2006](#); [Schulte et al., 2021](#)), and are associated with design choices supporting disassemblability and repairability ([MacNeill et al., 2020](#)). Across the reviewed studies, return timing, return rates, and the condition

of cores are linked to process stability and to the range of feasible remanufacturing actions.

Financial and operational constraints are also emphasized. Who bears the costs in CLSCs raises fairness concerns (Liu et al., 2022), while PSS models are described as offering predefined contractual structures that can influence this allocation. Interview evidence indicates substantial upfront investment is often required to initiate a remanufacturing venture (Liu et al., 2022). Several studies describe labor and scalability as practical factors: manual labor remains important for disassembly, scaling operations is associated with efficiency improvements (Goodall et al., 2014; Tamak et al., 2024), and batch processing is described as a means to reduce personnel costs (Andersen et al., 2024). These factors are presented as determinants of whether remanufacturing operations can be sustained beyond small scale.

Shared commitment is repeatedly described as a requirement for implementing remanufacturing at scale, but is complex due to the many interconnections across parties (Vogt Duberg et al., 2023). Boorsma et al. (2021) report that all departments must jointly commit to the process, describing a mindset shift across the company. Evidence also indicates that management should establish clear KPIs, while R&D develops operational processes for cleaning and testing (Benedettini, 2022; Boorsma et al., 2021). All involved departments should be trained for remanufacturing operations (Pagh Jensen et al., 2019), and incentive structures for sales staff may need adaptation so that selling remanufactured devices is rewarded similarly to new ones (Boorsma et al., 2021).

Perceptions of remanufactured devices are reported to vary across contexts (Hazen et al., 2015). In developing countries, stigma is described as less pronounced (Otur et al., 2021), and remanufacturing is reported as a strategy for increasing access to medical equipment (Eze et al., 2020). Rising technological capabilities are associated with higher acquisition costs, and remanufacturing is framed as a way to reduce financial barriers (Otur et al., 2021). At the same time, inhibitors are documented, including corruption, limited funding, and inadequate regulation (Eze et al., 2020; Samenjo et al., 2023), as well as weak legislation, shortages of skilled workers, and high upfront costs

(Otur et al., 2021). These conditions are described as affecting the ability to establish and maintain remanufacturing capacity, shaping supply-chain reliability and quality assurance.

3.3 Safety, Liability and Regulation

Terminology and definitions are described as inconsistent across jurisdictions, producing uncertainty about allowable practices. Some sources note that legislation may classify cores as waste (Gray & Charter, 2007); Parker et al. (2015, p. 53) explain this ambiguity may add "additional administrative and compliance costs to a business." This variability is described as being particularly challenging for smaller companies, while regulation is also presented as necessary to maintain product quality and safety (Shukla et al., 2023).

The medical device sector faces heightened constraints due to stringent hygiene and quality requirements, with device criticality determining the level of cleaning, validation, and documentation required (Benedettini, 2022; Schulte et al., 2021). The literature highlights a tradeoff between hygiene-driven trust and sustainability: some authors argue that standards are excessively escalated and can hinder remanufacturing adoption (Guzzo et al., 2020), citing patient safety concerns and liability risks as deterrents (Andersen et al., 2024; Parker et al., 2015). At the same time, restoration to "as-new" condition under strict safety and quality controls is emphasized as essential for trust and regulatory acceptance (Reinbold & Zaeh, 2025), underscoring how hygiene requirements shape both the feasibility and oversight of remanufacturing.

Regulation is frequently described as a deterrent when compliance makes remanufacturing less attractive than producing new devices (Baron, 2016). In the United States, requirements to demonstrate the safety of remanufactured devices are reported to incentivize single-use labeling, while OEM influence and stricter FDA rules further constrain remanufacturing relative to new production (Gray & Charter, 2007; MacNeill et al., 2020). At the same time, regulation can act as an enabler by increasing confidence and accountability: standards such as Good Refurbishment Practice (GRP) formalize safe refurbishment, and directives including ELV

and WEEE establish producer responsibility and clearer governance for end-of-use treatment (Benedettini, 2022; Plumeyer & Braun, 2011).

4 Discussion

Collectively, the reviewed studies indicate that the feasibility of remanufacturing across multiple lifecycles is primarily shaped by the coupling of product design, stable core recovery, and regulatory conditions, rather than by isolated technical limitations. Design features such as disassembly access, reversible connections, material durability, and cleanability repeatedly emerge as prerequisites for safe and economically viable remanufacturing (Gray & Charter, 2007; Hoveling et al., 2024; Ijomah et al., 2006; Kane et al., 2017). These characteristics enable targeted component replacement and upgrading, but only when embedded within supply chains that can reliably return products while sufficient residual value remains (Sierra-Fontalvo et al., 2024; Vogt Duberg et al., 2023). Across the literature, remanufacturing emerges as an interlinked system in which design for disassembly and modularity impact the feasibility and cost of operational processes, and thereby determine which business models can manage return flows and obsolescence, and sustain repeated use cycles over time (Asif et al., 2021; Gray & Charter, 2007).

A consistent pattern is the interdependence between technical design decisions and return logistics. Modular architectures and standardized components allow selective replacement and upgrading during remanufacturing, addressing different rates of technological and material obsolescence (Asif et al., 2021; Kane et al., 2017). However, these technical possibilities remain constrained by uncertainty in core timing and quality, which is repeatedly identified as a major operational risk (Benedettini, 2022; Goodall et al., 2014). Trade-in schemes and product-service systems are presented as mechanisms to reduce this uncertainty by structuring return cycles and improving core condition (Barquet et al., 2013; Fofou et al., 2021; Pagh Jensen et al., 2019). This reinforces the finding that upgrade-oriented remanufacturing depends as much on business model design as on product architecture.

The literature also reveals important contradictions, particularly in relation to medical device remanufacturing. While some authors argue that single-use devices should not be remanufactured due to safety and liability concerns (Plumeyer & Braun, 2011), others document technically successful remanufacturing of such devices under controlled conditions (Guo et al., 2014; Schulte et al., 2021). Similarly, hygiene and regulatory requirements are framed both as disproportionate barriers that suppress circular practices (Andersen et al., 2024; Guzzo et al., 2020) and as essential safeguards for trust and patient safety (Reinbold & Zaeh, 2025; Shukla et al., 2023). These tensions suggest that remanufacturing becomes most constrained when regulatory expectations are unclear, unevenly applied, or misaligned with actual risk, increasing administrative burden and uncertainty (Benedettini, 2022; Parker et al., 2015).

From a theoretical perspective, the findings suggest that modularity, predefined return cycles, and platform-based design function as deliberate strategies for controlling how and when obsolescence is addressed within regulatory constraints (Asif et al., 2021; Gray & Charter, 2007). Yet most existing analytical tools prioritize economic and environmental performance, offering only limited insight into how upgrading, social impacts, and regulatory activities are managed when product functionality changes (Leemans et al., 2025; Mejía-Moncayo et al., 2023). This leaves a structural gap in current frameworks, particularly for sectors where regulatory approval and validation are central to remanufacturing feasibility.

Future research should therefore move beyond generic remanufacturability assessments toward context-specific models that integrate upgradeability, uncertainty in core availability and condition, and the regulatory effort required when product functionality changes (Asif et al., 2021; Goodall et al., 2014). Empirical studies comparing trade-in and product-service system arrangements across regions would further clarify how business models influence the timing, quality, and upgrade potential of returned products (Benedettini, 2022; Oturu et al., 2021).

4.1 Limitations

This review is limited by its reliance on a single bibliographic (Google Scholar), English-language publications, and a relatively focused sample of the broader remanufacturing literature. As a result, region-specific regulatory practices, informal remanufacturing activities, and locally embedded operational strategies may be underrepresented, limiting the generalizability of the findings.

5 Conclusion

This literature review examined which operational constraints shape remanufacturing strategies for upgrading technical functionality in existing medical devices. The literature shows that such upgrading is only viable when design decisions such as disassembly access, modular architectures, material durability, and cleanability are aligned with predictable core return timing and sufficient residual value, and when regulatory requirements are clearly defined and consistently applied. While remanufacturing and circular design are well covered in general manufacturing research, far less attention is given to how functional upgrades interact with medical device regulation, hygiene validation, liability, and post market compliance. Existing remanufacturability and assessment tools largely prioritize economic or environmental performance, offering limited guidance on regulatory effort, functional change, and risk management. By combining these design, supply chain, and regulatory constraints, this review highlights specific gaps where healthcare focused decision tools and empirically grounded frameworks are still needed.

Declaration of Generative AI in Scientific Writing

During the preparation of this work the author used *ChatGPT 5.1* for finetuning the research question, and for generating search strings for Google Scholar.

ResearchRabbit was used to aid the storage and subsequent snowballing of sources.

ChatGPT 5.1 was used to cluster manually extracted quotes and topics from the papers, which were then used for writing the literature review.

ChatGPT 5.1 was then used to check for grammar mistakes, and enhance the clarity

and conciseness of the text. *ChatGPT 5.2 Thinking* was used to assist in rewriting the first draft.

After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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A Literature Review Protocol

A.1 Research Question

A systematic literature review was conducted based on the initial research question: *"What retrofit or remanufacturing strategies are documented for upgrading safety-critical functions in existing medical devices, and which technical or regulatory constraints most commonly shape their feasibility?"*

Throughout writing the literature review, the research question was changed to: *"What operational constraints are documented in remanufacturing strategies for upgrading technical functionality in existing medical devices?"*

A.2 Exclusion Criteria

To capture the development of the field while avoiding irrelevant or outdated legislation, the search date range was set from 2005 to the present day. Both English and Dutch literature were included, and no geographical restrictions were applied due to the scoping nature of this review.

A.3 Search Strings

ChatGPT 5.1 was used to generate a set of comprehensive and MECE (Mutually Exclusive, Collectively Exhaustive) search queries for Google Scholar, with explicit instructions to apply strategic wildcards. The resulting search strings were arranged from broad to increasingly specific:

1. ("medical device*" AND (retrofit* OR remanufactur* OR "design modific*" OR "post-market upgrade*")) AND ("safety-critical" OR "risk control*" OR "performance improvement") AND (MDR OR FDA OR "ISO 14971" OR "ISO 13485" OR "regulatory constraint*" OR "technical feasibility")
2. "medical device*" AND (retrofit* OR upgrade*) AND ("ISO 13485" OR "ISO 14971" OR IEC*)
3. "medical device*" AND ("design change*" OR modific*) AND (MDR OR "EU 2017/745" OR "Medical Device Regulation")
4. "medical device*" AND ("design change*" OR modific*) AND (FDA OR 510(k) OR "Quality System Regulation")
5. "medical device remanufacturing" AND (MDR OR "Medical Device Regulation" OR FDA OR "regulatory compliance")
6. "medical device*" AND remanufactur* AND ("lifecycle" OR "circular" OR "reuse")
7. "medical device*" AND (retrofit* OR "design change*") AND ("human factor*" OR usability OR workflow)
8. (retrofit* OR upgrade*) AND ("legacy system*" OR "embedded system*" OR "sensor integration") AND "medical device"
9. ("sensor*" OR "identification technology*" OR "barcode*" OR RFID OR "computer vision") AND ("medical device*" AND retrofit*)
10. "risk control*" AND ("medical device*" OR "health technology") AND (upgrade* OR modific*)
11. ("safety-critical" OR "safety function*") AND ("medical device*" OR "health-care equipment") AND (upgrade* OR retrofit*)
12. "post-market" AND ("modification*" OR upgrade* OR retrofit*) AND "medical device"
13. "medical device*" AND retrofit* AND ("upgrade*" OR "modernisation" OR "function* improvement")
14. "medical device*" AND retrofit* AND ("safety-critical" OR "safety function*" OR "risk control*")
15. remanufactur* AND "medical device*" AND ("safety" OR "performance upgrade*" OR "risk reduction")
16. (retrofit* OR "field upgrade*") AND ("automated medication" OR "drug dispens*" OR "medicine dispens*" OR "health* automation")

A.4 Retrieval and Screening

See Figure 1 for a graphical overview of the retrieval and screening process.

Google Scholar was used for each search string, with a lower date limit of 2005. The results were sorted by relevance, and the first 30 results for each string were screened based on their titles.

The accepted literature (78 papers) was imported into *ResearchRabbit*, where forward and backward citation tracking was used to expand the dataset in a 'snowballing' manner. This process yielded an initial pool of 169 papers.

A first manual filtering pass based on abstracts reduced the pool to 60 papers. A second manual filtering pass based on conclusions reduced this to 40 papers. Several were subsequently removed for insufficient relevance to remanufacturing, leaving 33 papers for evidence assessment through full-text reading.

17 papers were related to medical contexts, the remaining 16 papers covered either more general topics, or other case studies.

A.5 Evidence Assessment

Each of the 33 papers was examined using the *Strip* method described by Prof. David Stuckler (2021) to extract their core ideas. The reading order began with a general overview (headings, figures, and conclusion), proceeded to the key sections (introduction, methodology, and results), and concluded with all remaining sections (Prof. David Stuckler, 2022).

A.6 Analysis

All papers except two were manually read, and had their key sections pertaining to the research question extracted in a separate literature review document for clustering later on.

The works by Gray and Charter (2007) and Parker et al. (2015) had key snippets extracted by *ChatGPT 5.1* due to their length (70+ pages).

The resulting document with key sections (counting 38 pages) was used as input for *ChatGPT 5.1 (Thinking - Standard)* with the following prompt:

I am writing a literature review and have collected a document containing quotes, insights, and short notes extracted from multiple pa-

pers. Each new paper is headed by a Header 4 in blue, with the citation information. All the snippets in black are taken from that paper.

Your task is to perform a thematic synthesis of this document. Follow these steps:

1. Read all quotes carefully and assign initial codes representing the main idea of each quote.
 - Codes should be short (1-5 words).
 - A quote may receive more than one code.
2. Cluster similar codes into higher-order themes.
 - Produce 3-5 major themes.
 - For each theme, include 2-5 subthemes if appropriate.
3. For each major theme, list:
 - A clear theme title that reflects the concept.
 - A concise explanation of the theme.
 - The codes that cluster into this theme.
 - All supporting quotes, each with its source.
4. Ensure each theme is analytically useful for writing a literature review, meaning:
 - Themes should not be based on single papers unless absolutely necessary.
 - Themes should highlight patterns, divergences, contradictions, and gaps.
 - Themes should be directly relevant to understanding retrofit and remanufacturing strategies for medical devices and the constraints shaping feasibility.
5. After presenting the themes, provide:
 - A summary of how these themes relate to each other, and

- Suggested section headers I could use in the literature review.

Take the full quote document (attached as a PDF) and begin the thematic analysis.

The result was the identification of four themes, each with their core idea explained, a list of sub-themes, the codes related to the theme, and all relevant key sections sorted by source.

A.7 Synthesis

Based on the identified themes and clustered quotes, the first draft was manually written and passed through *ChatGPT 5.1* for actionable rewrite feedback. This draft was handed in for preliminary feedback.

A.8 Rewrite

Based on the feedback, the research question was altered along with the overall structure of the review. The four identified themes were consolidated into three, and the resulting manuscript was edited multiple times to sharpen argumentation.

Figure 1

Flow Diagram of the Literature Retrieval and Screening Process

